Date	Revision	Tag	Description
Nov. ?, 2013	0	rev0	Schematic capture complete. Preliminary placement.

Date	Revision	Tag	Description

Title: Revision History		
Size: Letter	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 1 of 17
File: C:\Users\..\Preamble_History.SchDoc		Drawn By: Adam Gilbert

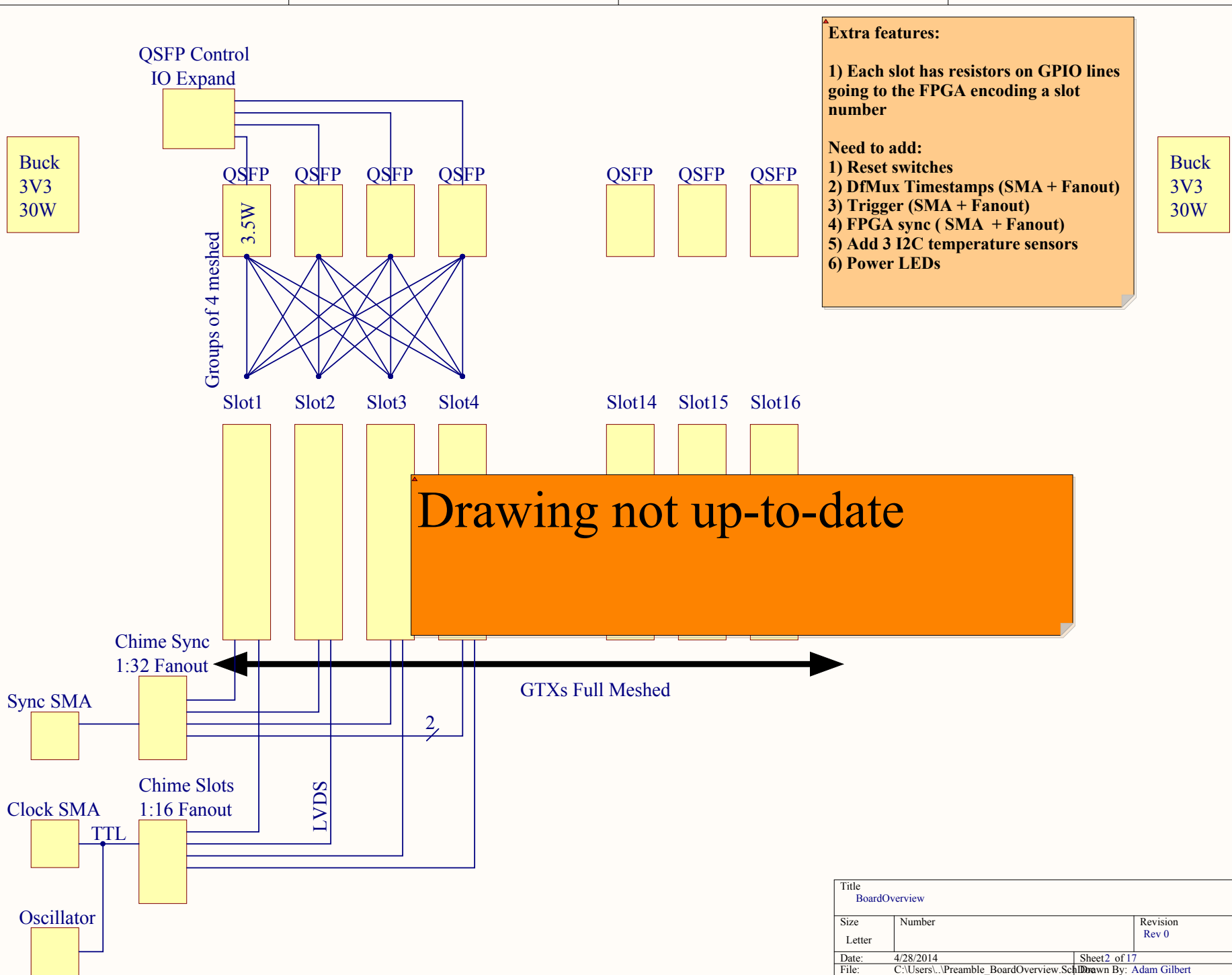

QSFP Control
IO Expand
Buck
3V3
30W
QSFP
QSFP
QSFP
QSFP
3.5W
Groups of 4 meshed
QSFP
QSFP
QSFP
Slot1
Slot2
Slot3
Slot4
Slot14
Slot15
Slot16
Extra features:
1) Each slot has resistors on GPIO lines going to the FPGA encoding a slot number
Need to add:
1) Reset switches
2) DfMux Timestamps (SMA + Fanout)
3) Trigger (SMA + Fanout)
4) FPGA sync (SMA + Fanout)
5) Add 3 I2C temperature sensors
6) Power LEDs
Buck
3V3
30W
Drawing not up-to-date
Chime Sync
1:32 Fanout
GTXs Full Meshed
Sync SMA
2
Chime Slots
1:16 Fanout
Clock SMA
TTL
LVDS
Oscillator
Title
BoardOverview
Size
Letter
Number
Revision
Rev 0
Date:
4/28/2014
Sheet2 of 17
File:
C:\Users\..\Preamble_BoardOverview.SchDoc
Drawn By:
Adam Gilbert

Diagram not pretty yet - sorry!

MONI2C - all slots bussed - I2C Matrix

EEPROM
I2C address 0b101.0111 (0x57)

TempSense x 2
I2C address 0b1001 110 (0x9C)
I2C address 0b1001 101 (0x9A)

QSFPControl
Slots 1-8: I2C address 0100 000b (0x40)
Slots 9-16: I2C address 0100 001b (0x42)

Did not verify if we have address collisions on Motherboard

MONI2C can be connected to POWI2C

All device addresses adjustabe

POWI2C - all slots bussed (BP1)

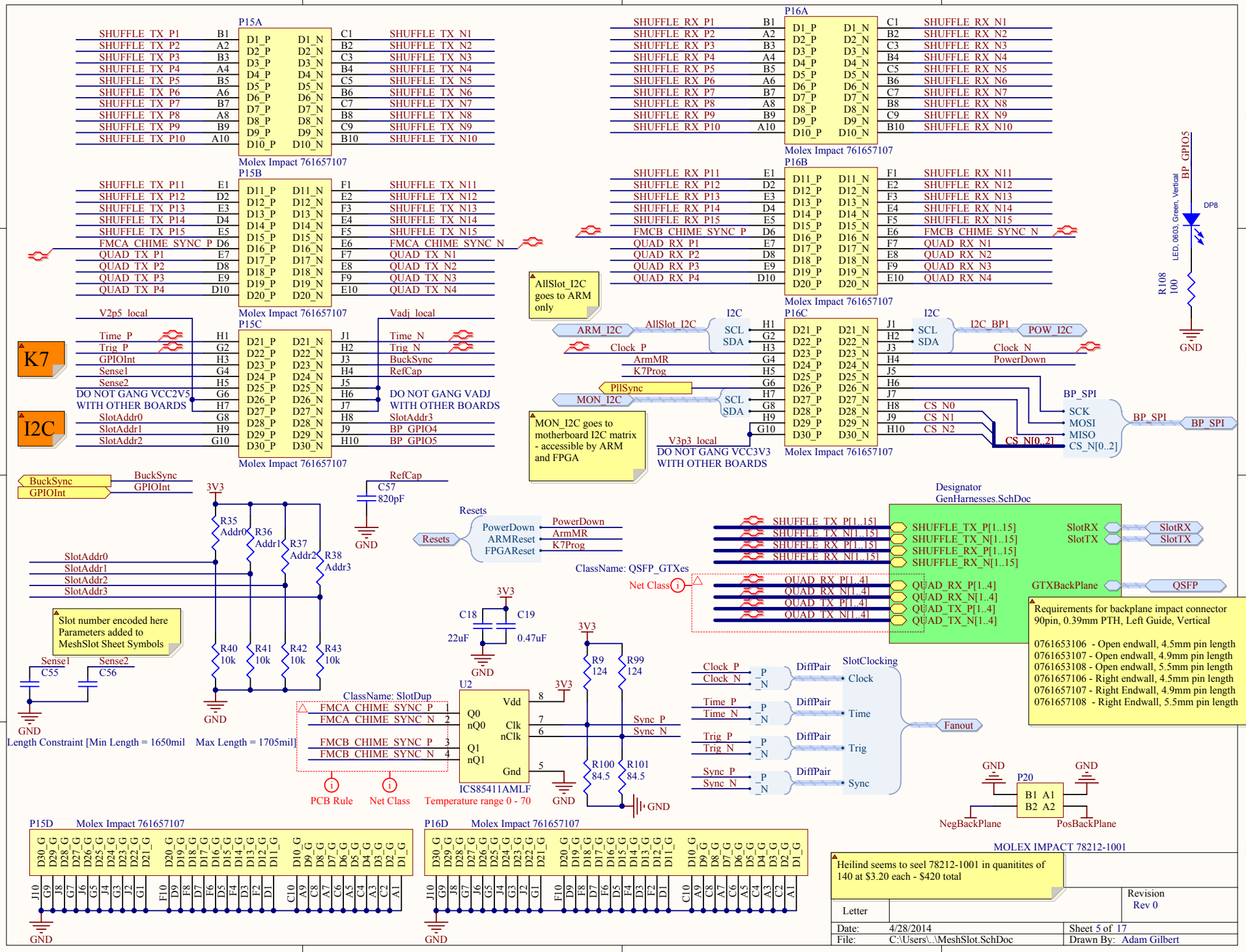
ResetControl
0b0100 010 (0x44)

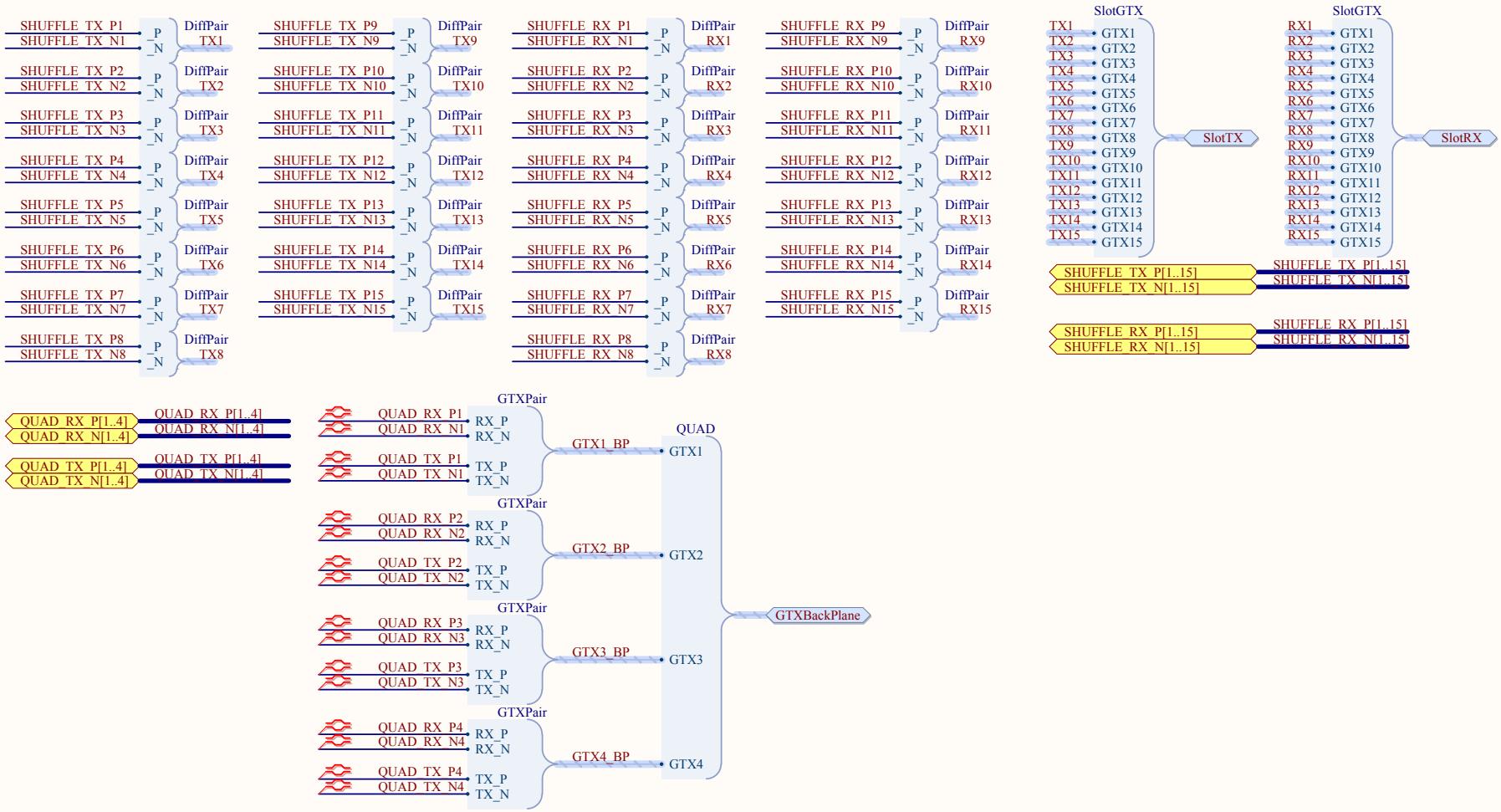
ARMI2C

All Arms bussed together

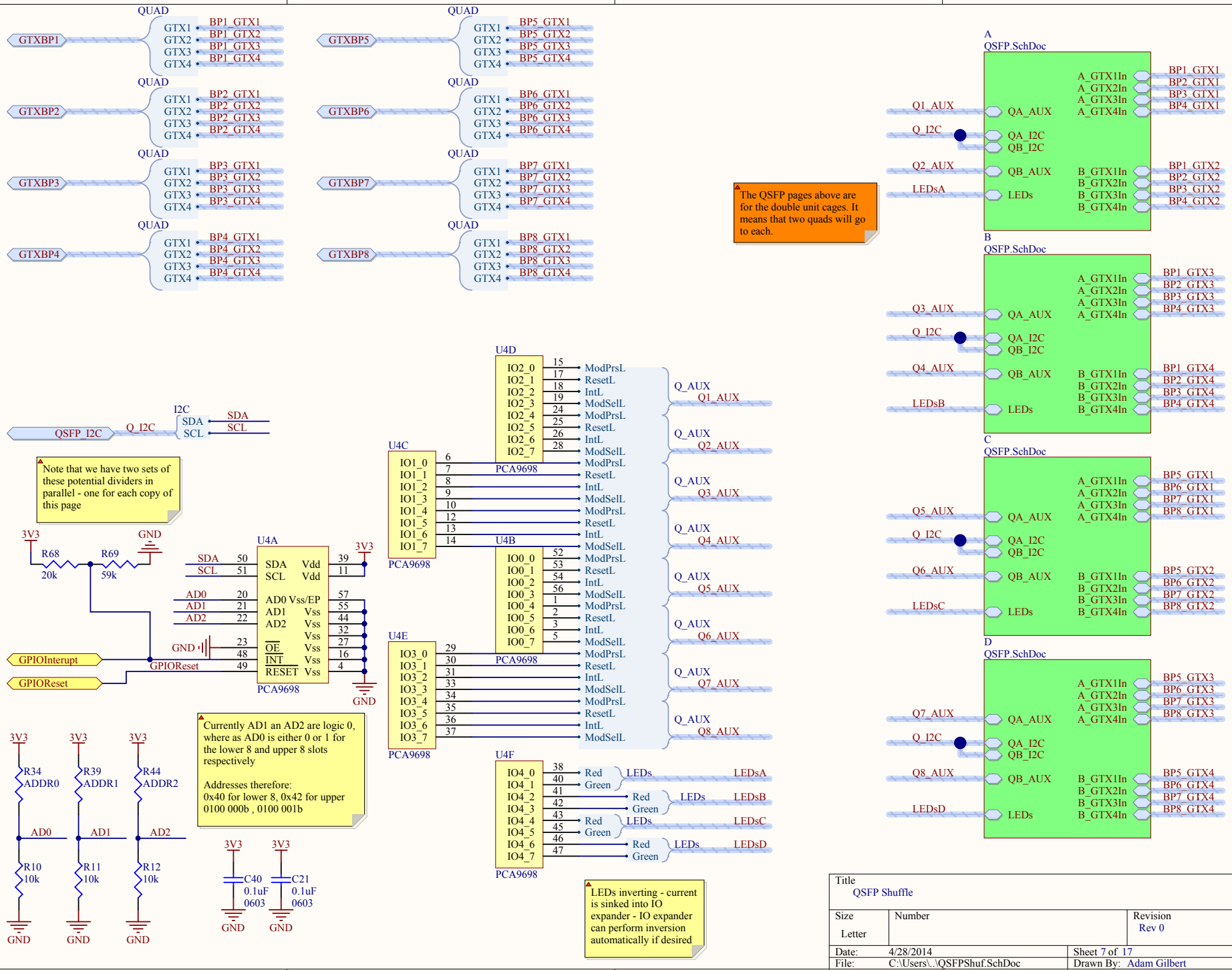
Title: Addressing		
Size: Letter	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 3 of 17
File: C:\Users\..\Preamble_Addressing.SchDoc		Drawn By: Adam Gilbert

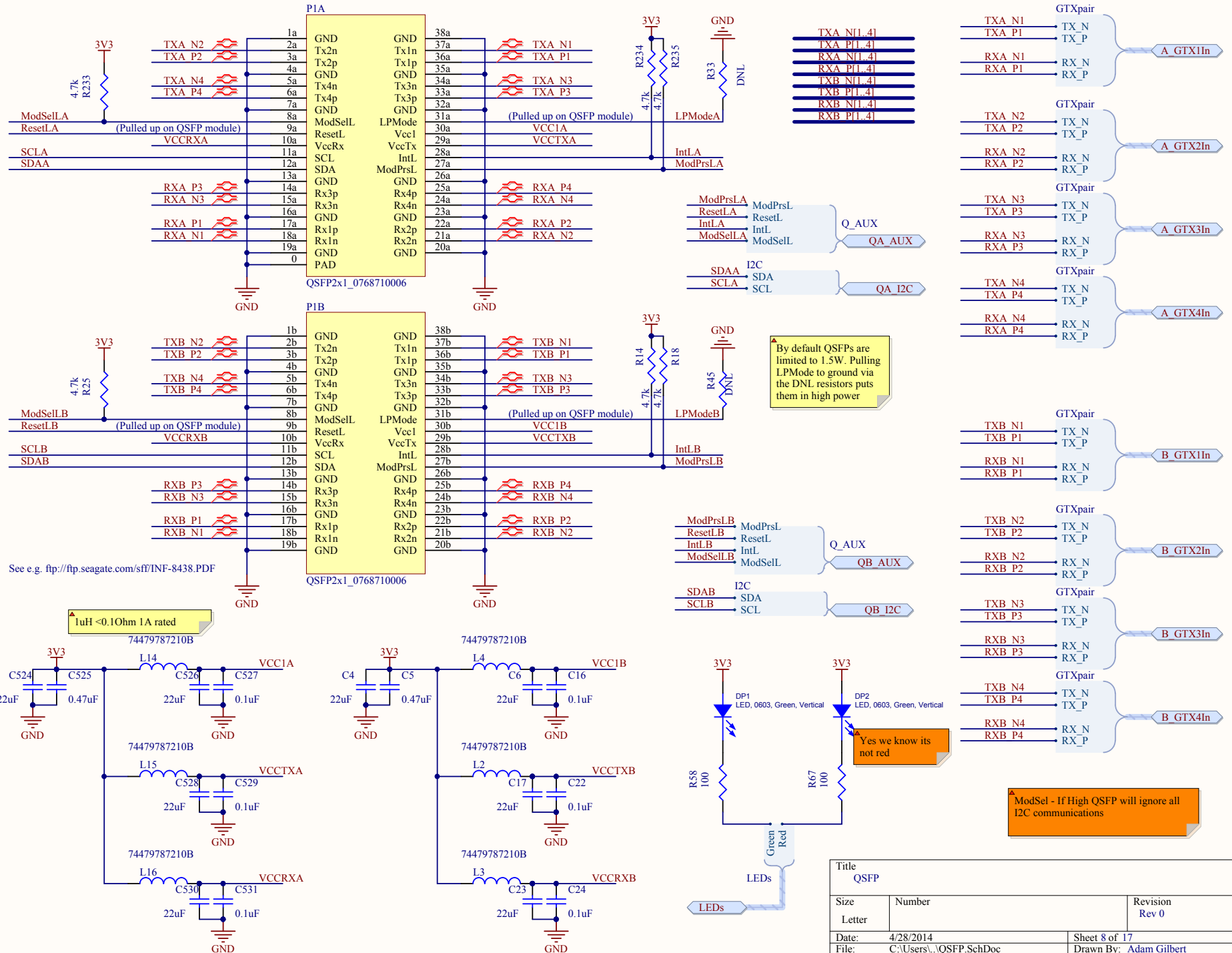

DNL=logic 0
0 = logic 1
Addr0: DNL Cap1: 820pF
Addr1: DNL Cap2: 820pF
Addr2: DNL
Addr3: DNL
S1
S2
S3
S4
Addr0: 0 Cap1: 820pF
Addr1: DNL Cap2: 2.2nF
Addr2: DNL
Addr3: DNL
Addr0: DNL Cap1: 820pF
Addr1: 0 Cap2: 3.9nF
Addr2: DNL
Addr3: DNL
Addr0: 0 Cap1: 820pF
Addr1: 0 Cap2: 5.6nF
Addr2: DNL
Addr3: DNL
Addr0: DNL Cap1: 2.2nF
Addr1: DNL Cap2: 820pF
Addr2: 0
Addr3: DNL
Addr0: 0 Cap1: 2.2nF
Addr1: DNL Cap2: 2.2nF
Addr2: 0
Addr3: DNL
Addr0: DNL Cap1: 2.2nF
Addr1: 0 Cap2: 3.9nF
Addr2: 0
Addr3: DNL
Addr0: 0 Cap1: 2.2nF
Addr1: 0 Cap2: 5.6nF
Addr2: 0
Addr3: DNL
S5
S6
S7
S8
TxSlot1
RxSlot1
TxSlot2
RxSlot2
TxSlot3
RxSlot3
TxSlot4
RxSlot4
TxSlot5
RxSlot5
TxSlot6
RxSlot6
TxSlot7
RxSlot7
TxSlot8
RxSlot8
SlotTX
SlotRX
GTXS1
GTXS2
GTXS3
GTXS4
GTXS5
GTXS6
GTXS7
GTXS8
QSFP
FanS1
FanS2
FanS3
FanS4
FanS5
FanS6
FanS7
FanS8
Fanout
BuckSync
ResS1
ResS2
ResS3
ResS4
ResS5
ResS6
ResS7
ResS8
PllSync
Resets
BP_SPI
MON_I2C
POW_I2C
ARM_I2C
GPIOInt
MeshSlot.SchDoc
What do we do with BP_SPI and PLLSync?
Addr0: DNL Cap1: 3.9nF
Addr1: DNL Cap2: 820pF
Addr2: DNL
Addr3: 0
Addr0: 0 Cap1: 3.9nF
Addr1: DNL Cap2: 2.2nF
Addr2: DNL
Addr3: 0
Addr0: DNL Cap1: 3.9nF
Addr1: 0 Cap2: 3.9nF
Addr2: DNL
Addr3: 0
Addr0: 0 Cap1: 3.9nF
Addr1: 0 Cap2: 5.6nF
Addr2: DNL
Addr3: 0
Addr0: DNL Cap1: 5.6nF
Addr1: DNL Cap2: 820pF
Addr2: 0
Addr3: 0
Addr0: 0 Cap1: 5.6nF
Addr1: DNL Cap2: 2.2nF
Addr2: 0
Addr3: 0
Addr0: DNL Cap1: 5.6nF
Addr1: 0 Cap2: 3.9nF
Addr2: 0
Addr3: 0
Addr0: 0 Cap1: 5.6nF
Addr1: 0 Cap2: 5.6nF
Addr2: 0
Addr3: 0
S9
S10
S11
S12
S13
S14
S15
S16
TxSlot9
RxSlot9
TxSlot10
RxSlot10
TxSlot11
RxSlot11
TxSlot12
RxSlot12
TxSlot13
RxSlot13
TxSlot14
RxSlot14
TxSlot15
RxSlot15
TxSlot16
RxSlot16
GTXS9
GTXS10
GTXS11
GTXS12
GTXS13
GTXS14
GTXS15
GTXS16
FanS9
FanS10
FanS11
FanS12
FanS13
FanS14
FanS15
FanS16
ResS9
ResS10
ResS11
ResS12
ResS13
ResS14
ResS15
ResS16
GPIOInt
GPIOReset
QSFPShuf.SchDoc
QSha
QShb
GPIOInterupt
QSFP_I2C
GTXBP1
GTXBP2
GTXBP3
GTXBP4
GTXBP5
GTXBP6
GTXBP7
GTXBP8
ADDR1: DNL
ADDR0: DNL
ADDR2: DNL
ADDR1: 0
ADDR0: DNL
ADDR2: DNL
Mesh
FullMesh.SchDoc
Fanouts
SlotClockTimingSyncTrig.SchDoc
S1Clocking
S2Clocking
S3Clocking
S4Clocking
S5Clocking
S6Clocking
S7Clocking
S8Clocking
S9Clocking
S10Clocking
S11Clocking
S12Clocking
S13Clocking
S14Clocking
S15Clocking
S16Clocking
Osc
Oscillator.SchDoc
Mech
Mechanical.SchDoc
Resets
Resets.SchDoc
EEP
ExtraI2C.SchDoc
PWR
BuckRegulator
Title
Size
Letter
Number
Revision
Date: 4/28/2014
File: C:\Users\..\Slots1To16.SchDoc
Sheet of
Drawn By:

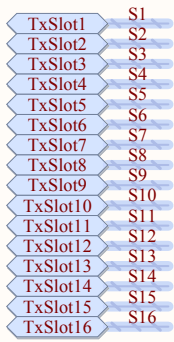

P15A
P15B
P15C
P16A
P16B
P16C
Molex Impact 761657107
K7
I2C
AllSlot_I2C goes to ARM only
MON_I2C goes to motherboard I2C matrix - accessible by ARM and FPGA
DO NOT GANG VCC2V5 WITH OTHER BOARDS
DO NOT GANG VADJ WITH OTHER BOARDS
DO NOT GANG VCC3V3 WITH OTHER BOARDS
BP_GPIO5
DP8
LED, 0603, Green, Vertical
R108 100
GND
BuckSync
GPIOInt
RefCap
C57 820pF
Resets
PowerDown
ARMReset
FPGAReset
ArmMR
K7Prog
3V3
R35 Addr0
R36 Addr1
R37 Addr2
R38 Addr3
SlotAddr0
SlotAddr1
SlotAddr2
SlotAddr3
Slot number encoded here
Parameters added to
MeshSlot Sheet Symbols
Sense1 C55
Sense2 C56
R40 10k
R41 10k
R42 10k
R43 10k
Length Constraint [Min Length = 1650mil
Max Length = 1705mil]
C18 22uF
C19 0.47uF
U2
ICS85411AMLF
Temperature range 0 - 70
ClassName: SlotDup
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
PCB Rule
Net Class
R9 124
R99 124
R100 84.5
R101 84.5
Sync_P
Sync_N
ClassName: QSFP_GTXes
Net Class
Designator
GenHarnesses.SchDoc
SHUFFLE_TX_P[1..15]
SHUFFLE_TX_N[1..15]
SHUFFLE_RX_P[1..15]
SHUFFLE_RX_N[1..15]
QUAD_RX_P[1..4]
QUAD_RX_N[1..4]
QUAD_TX_P[1..4]
QUAD_TX_N[1..4]
SlotRX
SlotTX
GTXBackPlane
QSFP
Requirements for backplane impact connector
90pin, 0.39mm PTH, Left Guide, Vertical
0761653106 - Open endwall, 4.5mm pin length
0761653107 - Open endwall, 4.9mm pin length
0761653108 - Open endwall, 5.5mm pin length
0761657106 - Right endwall, 4.5mm pin length
0761657107 - Right Endwall, 4.9mm pin length
0761657108 - Right Endwall, 5.5mm pin length
Clock_P
Clock_N
Time_P
Time_N
Trig_P
Trig_N
Sync_P
Sync_N
DiffPair
SlotClocking
Clock
Time
Trig
Sync
Fanout
P20
B1 A1
B2 A2
NegBackPlane
PosBackPlane
MOLEX IMPACT 78212-1001
Heilind seems to seel 78212-1001 in quanitites of 140 at $3.20 each - $420 total
P15D
P16D
Letter
Revision
Rev 0
Date: 4/28/2014
Sheet 5 of 17
File: C:\Users\..\MeshSlot.SchDoc
Drawn By: Adam Gilbert



Title: Generate Harnesses		
Size: Letter	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 6 of 17
File: C:\Users\..\GenHarnesses.SchDoc		Drawn By: Adam Gilbert

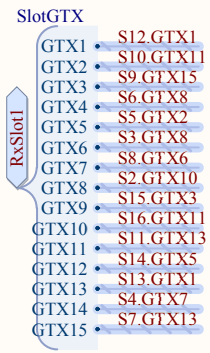

QUAD
GTXBP1
GTX1 BP1_GTX1
GTX2 BP1_GTX2
GTX3 BP1_GTX3
GTX4 BP1_GTX4
GTXBP2
BP2_GTX1 BP2_GTX2 BP2_GTX3 BP2_GTX4
GTXBP3
BP3_GTX1 BP3_GTX2 BP3_GTX3 BP3_GTX4
GTXBP4
BP4_GTX1 BP4_GTX2 BP4_GTX3 BP4_GTX4
GTXBP5
BP5_GTX1 BP5_GTX2 BP5_GTX3 BP5_GTX4
GTXBP6
BP6_GTX1 BP6_GTX2 BP6_GTX3 BP6_GTX4
GTXBP7
BP7_GTX1 BP7_GTX2 BP7_GTX3 BP7_GTX4
GTXBP8
BP8_GTX1 BP8_GTX2 BP8_GTX3 BP8_GTX4
The QSFP pages above are for the double unit cages. It means that two quads will go to each.
A
QSFP.SchDoc
A_GTX1In A_GTX2In A_GTX3In A_GTX4In
Q1_AUX QA_AUX
Q_I2C QA_I2C QB_I2C
Q2_AUX QB_AUX
LEDsA LEDs
B_GTX1In B_GTX2In B_GTX3In B_GTX4In
BP1_GTX1 BP2_GTX1 BP3_GTX1 BP4_GTX1
BP1_GTX2 BP2_GTX2 BP3_GTX2 BP4_GTX2
B
QSFP.SchDoc
Q3_AUX
Q4_AUX
LEDsB
BP1_GTX3 BP2_GTX3 BP3_GTX3 BP4_GTX3
BP1_GTX4 BP2_GTX4 BP3_GTX4 BP4_GTX4
C
QSFP.SchDoc
Q5_AUX
Q6_AUX
LEDsC
BP5_GTX1 BP6_GTX1 BP7_GTX1 BP8_GTX1
BP5_GTX2 BP6_GTX2 BP7_GTX2 BP8_GTX2
D
QSFP.SchDoc
Q7_AUX
Q8_AUX
LEDsD
BP5_GTX3 BP6_GTX3 BP7_GTX3 BP8_GTX3
BP5_GTX4 BP6_GTX4 BP7_GTX4 BP8_GTX4
QSFP_I2C
Q_I2C
I2C
SDA SCL
Note that we have two sets of these potential dividers in parallel - one for each copy of this page
3V3
R68 20k
R69 59k
GND
U4A
PCA9698
AD0 AD1 AD2
GPIOInterupt
GPIOReset
Currently AD1 an AD2 are logic 0, where as AD0 is either 0 or 1 for the lower 8 and upper 8 slots respectively
Addresses therefore:
0x40 for lower 8, 0x42 for upper
0100 000b , 0100 001b
R34 ADDR0
R39 ADDR1
R44 ADDR2
R10 10k
R11 10k
R12 10k
C40 0.1uF 0603
C21 0.1uF 0603
U4D
U4C
U4B
U4E
U4F
IO2_0 IO2_1 IO2_2 IO2_3 IO2_4 IO2_5 IO2_6 IO2_7
IO1_0 IO1_1 IO1_2 IO1_3 IO1_4 IO1_5 IO1_6 IO1_7
IO0_0 IO0_1 IO0_2 IO0_3 IO0_4 IO0_5 IO0_6 IO0_7
IO3_0 IO3_1 IO3_2 IO3_3 IO3_4 IO3_5 IO3_6 IO3_7
IO4_0 IO4_1 IO4_2 IO4_3 IO4_4 IO4_5 IO4_6 IO4_7
ModPrsL ResetL IntL ModSelL ModPrsL ResetL IntL ModSelL
Q_AUX Q1_AUX
Q_AUX Q2_AUX
Q_AUX Q3_AUX
Q_AUX Q4_AUX
Q_AUX Q5_AUX
Q_AUX Q6_AUX
Q_AUX Q7_AUX
Q_AUX Q8_AUX
Red Green
LEDs
LEDsA LEDsB LEDsC LEDsD
LEDs inverting - current is sinked into IO expander - IO expander can perform inversion automatically if desired
Title
QSFP Shuffle
Size
Letter
Number
Revision
Rev 0
Date: 4/28/2014
File: C:\Users\..\QSFPShuf.SchDoc
Sheet 7 of 17
Drawn By: Adam Gilbert

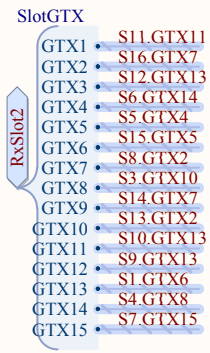

P1A
QSFP2x1_0768710006
P1B
QSFP2x1_0768710006
3V3
GND
TXA_N2
TXA_P2
TXA_N4
TXA_P4
TXA_N1
TXA_P1
TXA_N3
TXA_P3
RXA_P3
RXA_N3
RXA_P1
RXA_N1
RXA_P4
RXA_N4
RXA_P2
RXA_N2
4.7k R233
ModSelLA
ResetLA
(Pulled up on QSFP module)
VCCRXA
SCLA
SDAA
VCC1A
VCCTXA
R234 4.7k
R235 4.7k
R33 DNL
LPModeA
IntLA
ModPrsLA
TXB_N2
TXB_P2
TXB_N4
TXB_P4
TXB_N1
TXB_P1
TXB_N3
TXB_P3
RXB_P3
RXB_N3
RXB_P1
RXB_N1
RXB_P4
RXB_N4
RXB_P2
RXB_N2
4.7k R25
ModSelLB
ResetLB
VCCRXB
SCLB
SDAB
VCC1B
VCCTXB
R14 4.7k
R18 4.7k
R45 DNL
LPModeB
IntLB
ModPrsLB
See e.g. ftp://ftp.seagate.com/sff/INF-8438.PDF
By default QSFPs are limited to 1.5W. Pulling LPMode to ground via the DNL resistors puts them in high power
TXA_N[1..4]
TXA_P[1..4]
RXA_N[1..4]
RXA_P[1..4]
TXB_N[1..4]
TXB_P[1..4]
RXB_N[1..4]
RXB_P[1..4]
ModPrsLA
ResetLA
IntLA
ModSelLA
ModPrsL
ResetL
IntL
ModSelL
Q_AUX
QA_AUX
SDAA
SCLA
I2C
SDA
SCL
QA_I2C
ModPrsLB
ResetLB
IntLB
ModSelLB
QB_AUX
SDAB
SCLB
QB_I2C
GTXpair
TX_N
TX_P
RX_N
RX_P
A_GTX1In
A_GTX2In
A_GTX3In
A_GTX4In
B_GTX1In
B_GTX2In
B_GTX3In
B_GTX4In
1uH <0.1Ohm 1A rated
74479787210B
L14
L15
L16
L4
L2
L3
C524
C525
22uF
0.47uF
C526
C527
0.1uF
C528
C529
C530
C531
C4
C5
C6
C16
C17
C22
C23
C24
VCC1A
VCCTXA
VCCRXA
VCC1B
VCCTXB
VCCRXB
DP1
LED, 0603, Green, Vertical
DP2
LED, 0603, Green, Vertical
Yes we know its not red
R58 100
R67 100
Green
Red
LEDs
ModSel - If High QSFP will ignore all I2C communications
Title
QSFP
Size
Letter
Number
Revision
Rev 0
Date:
4/28/2014
Sheet 8 of 17
File:
C:\Users\..\QSFP.SchDoc
Drawn By:
Adam Gilbert

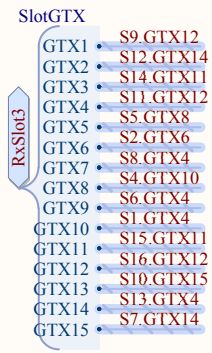


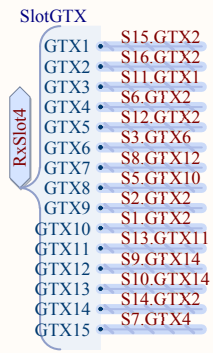
For checking purposes - you do not care about the GTX number in the netlabels, neither do you care about the ordering

As an example, on Slot 1, you must have S2 to S16 connected to it (Slot 1 is naturally connected)

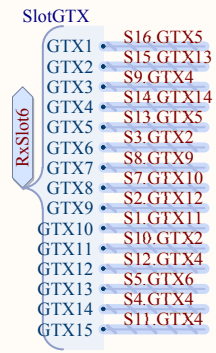


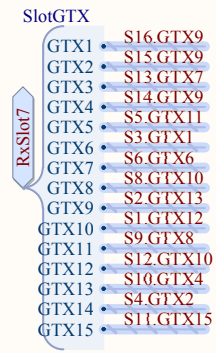


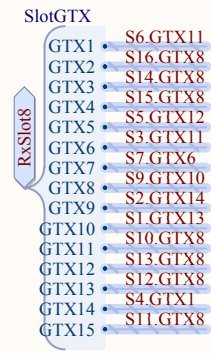


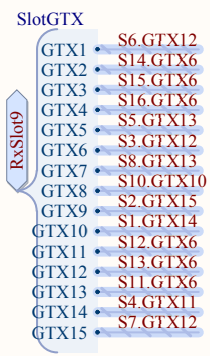




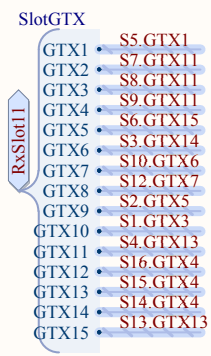


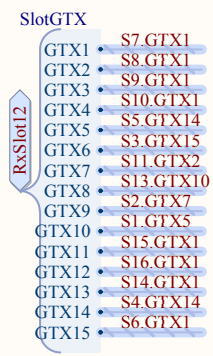


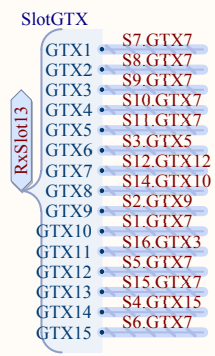


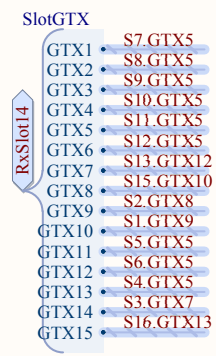


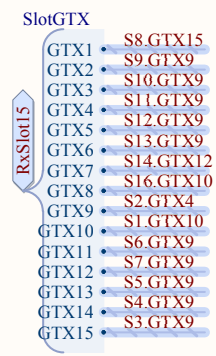










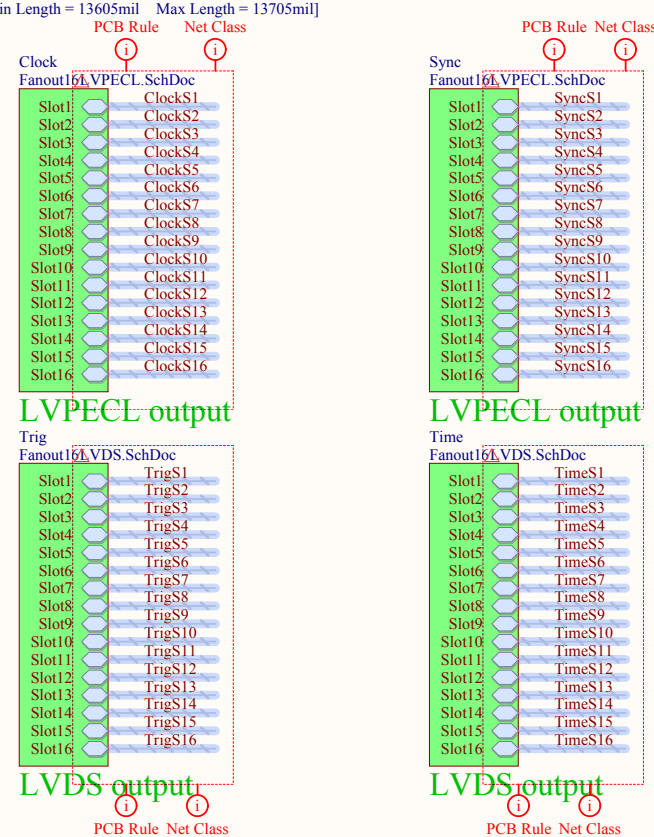


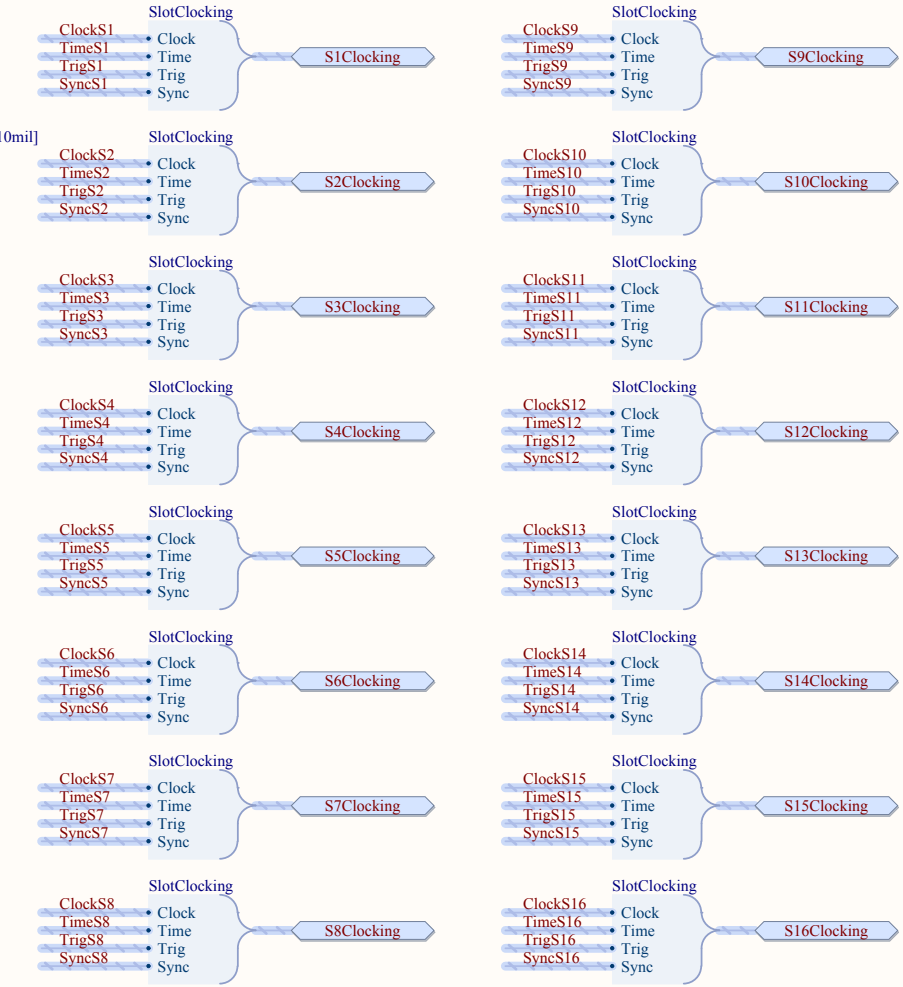


Title: FullMesh		
Size: A	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 10 of 17
File: C:\Users\..\FullMesh.SchDoc		Drawn By: Adam Gilbert

ClassName: SlotClocks
Length Constraint [Min Length = 13605mil Max Length = 13705mil]

ClassName: SlotSyncs
Length Constraint [Min Length = 11510mil Max Length = 11610mil]

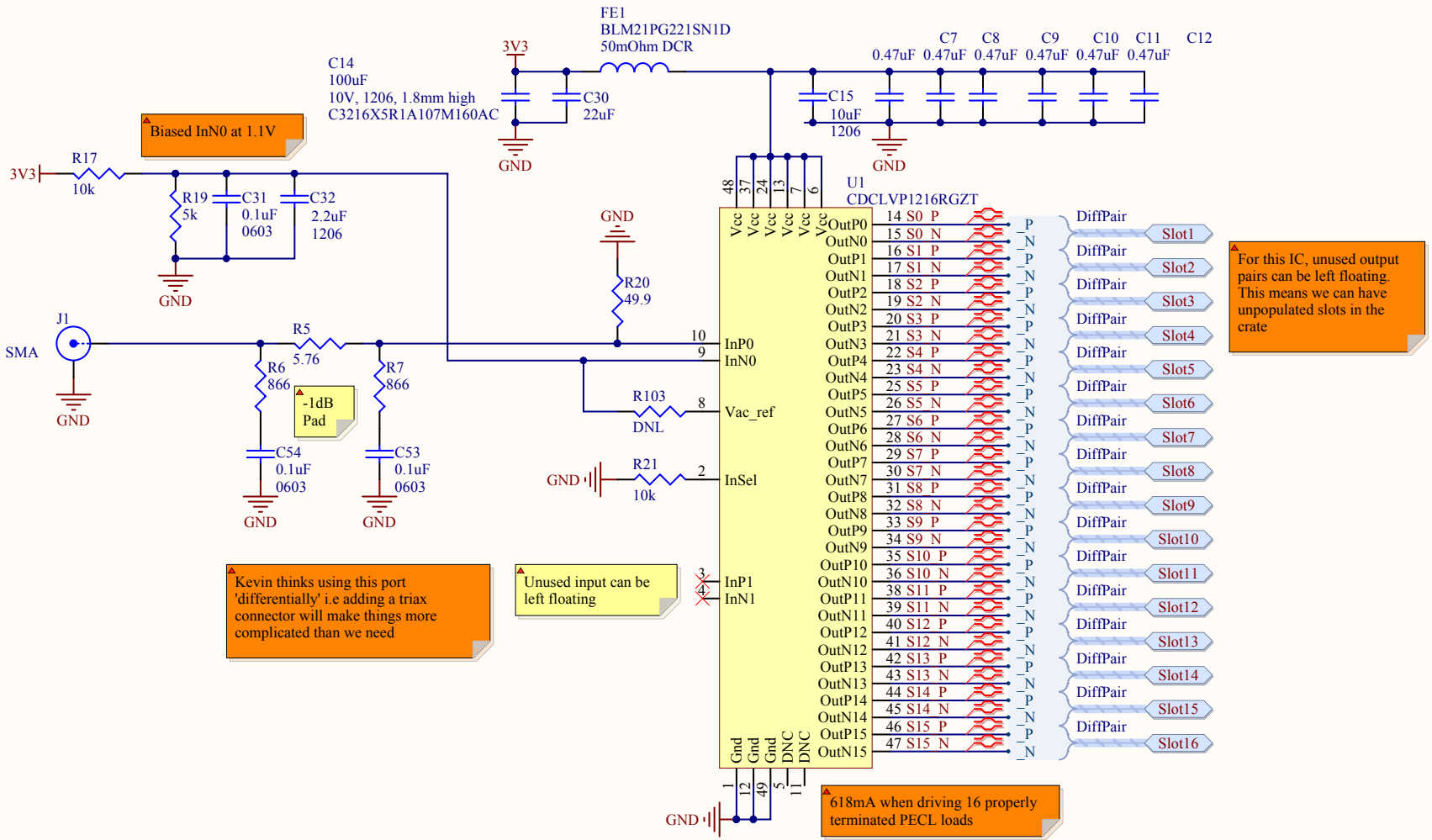




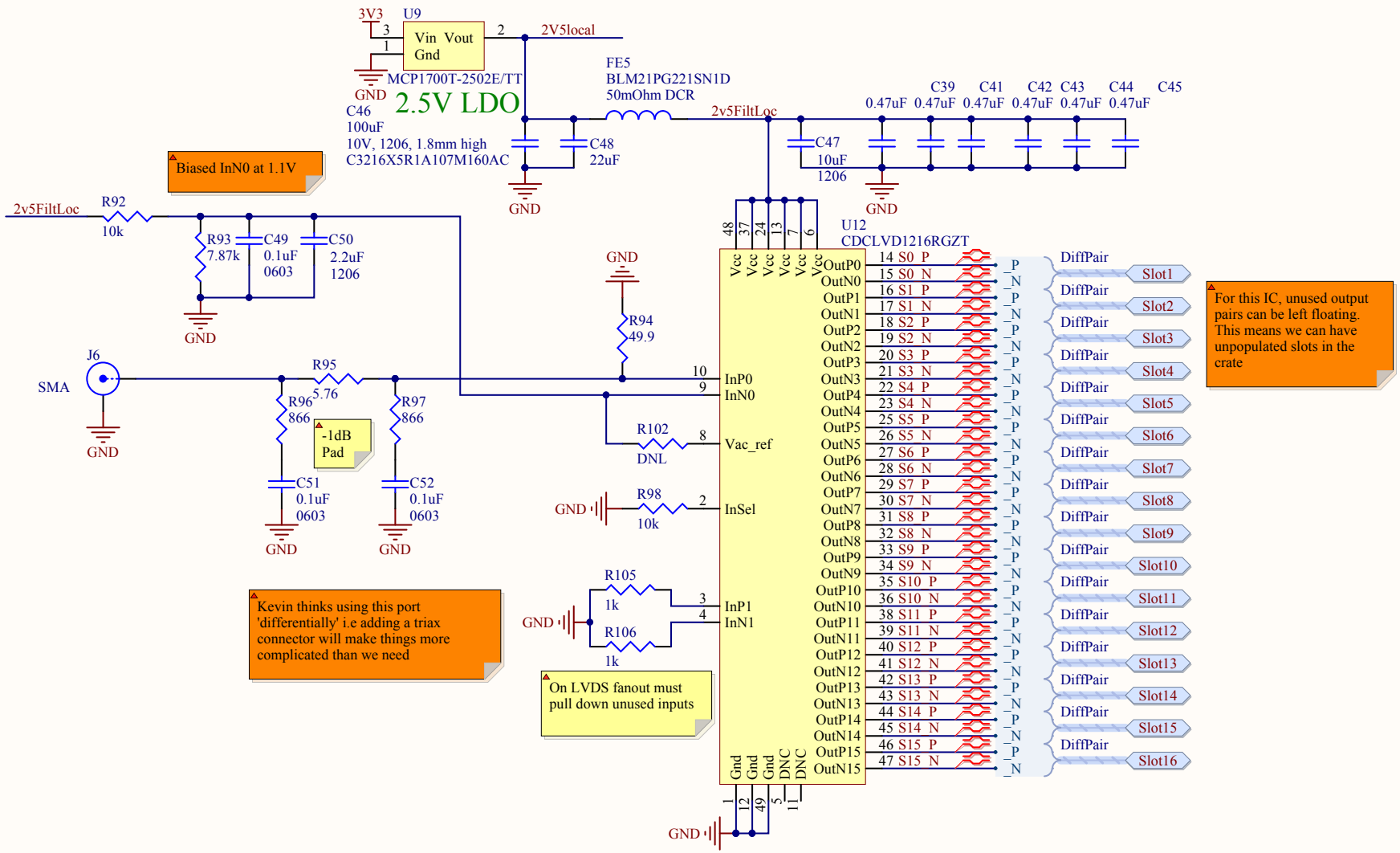
ClassName: SlotTrigs
Length Constraint [Min Length = 9955mil Max Length = 10055mil]

ClassName: SlotTime
Length Constraint [Min Length = 11555mil Max Length = 11655mil]

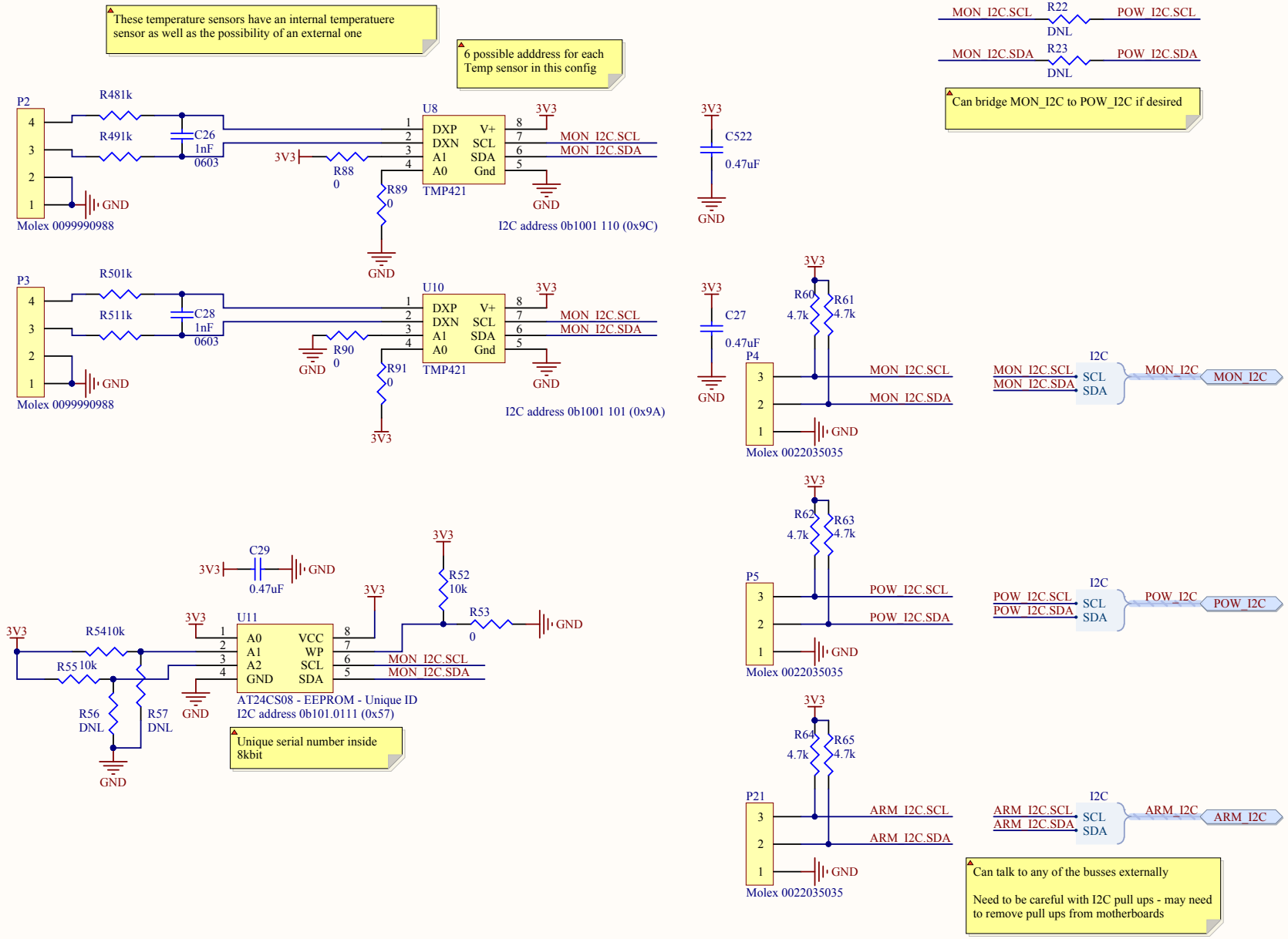
Title: Replicating Clock, Timing, Sync, and Trig signals for each Slot		
Size: Letter	Number	Revision: Rev 0
Date: 4/28/2014		Sheet 11 of 17
File: C:\Users\..\SlotClockTimingSyncTrig.SchDoc		Drawn By: Adam Gilbert



Title: 1:16 Clock Fanout		
Size: A	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 12 of 17
File: C:\Users\..\Fanout16LVPECL.SchDoc		Drawn By: Adam Gilbert



Title: 1:16 Clock Fanout		
Size: A	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 12 of 17
File: C:\Users\..\Fanout16LVDS.SchDoc		Drawn By: Adam Gilbert



Title: ExtraI2C		
Size: Letter	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 13 of 17
File: C:\Users\..\ExtraI2C.SchDoc		Drawn By: Adam Gilbert

I think the pulse will last 100mS, wanted it long enough to notice the LED flash p.6 of http://www.ti.com/lit/ds/symlink/sn74lvc1g123.pdf implies its okay to have a large cap like this

GPIO is open drain, and by default pins are inputs - pull ups ensure no reset asserted

Saved a package by using diode in package above

Molex 74099-1004

Title	Resets	
Size: Letter	Number	Revision: Rev 0
Date: 4/28/2014	Sheet 14 of 17	
File: C:\Users\..\Resets.SchDoc	Drawn By: Adam Gilbert	

Voltage inbetween AC coupling capacitor and diode in the below circuit always tends to 0 when no switching is occuring. When switching present, voltage is not allowed to become negative.

The buck sync sees things in the following order:
a) Inverter
b) Clamped AC coupling with 0 default
c) Inverter Inverter - giving 0 default - only needed the last two inverters to ensure strong drive currents to the buck

PosBackPlane 1.8A @ 16V in

4x C446 C447 C448 C449
10uF 1206
GMK316BJ106KL-T

If Internal regulator is to be used (for gate drive voltage), tie Vin port to PVin port and let VCC/LDO port float

PosBackPlane C2 10uF 1206 GMK316BJ106KL-T
C218 0.1uF 0402

3V3
C219 0.1uF 0402
C635 0.1uF 0402
C630 0.1uF 0402

9A
91%

TP1 1.8uH, 2.6mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high TP2
L1 SER1360-182

3V3
3x 100uF 10V, 1206 C3216X5R1A107M160AC
C444 C220 C221

BuckSync
VCC3V3
R13 100k
R215 100k
D9 Schottky Diode CDBU00340
C3 0.1uF 0402

U65
1A 1Y 2A 2Y 3A 3Y Gnd Vcc
SN74LVC3G14DCTR
R104 300k
VCC3V3

C13 0.1uF 0402

DP9 LED, 0603, Green, Vertical
R109 301
BuckAnalogGnd C233 1nF, 0603

U14 IR3895
S_Ctrl Vref Vin PVin Boot
Vcc/LDO_out SW
PGood
Vp
Sense
Rt/Sync
Enable Gnd Gnd PGnd Comp FB

Cboot1 0.1uF 0402
VSw
R139 20
RsA1 19.1k
RfbA1 19.1k
CcA1 2.2nF 0603
RcA1 127

C234 2.2uF 1206
R15 1k
BuckPowerGood

Over-voltage output protection
Analog Feedback

D3 Schottky Diode CDBU00340
Rt1 23.2k
Rt = 23.2k for 1Mhz free-run

RsB1 3.4k
RfbB1 3.4k
BuckAnalogGnd

If Vsense exceeds 1.2*Vp an over-voltage protection is initiated. Note that choice of RseqA, RseqB and the choice of Master voltage can effect the trip level

3V3
DP10 LED, 0603, Green, Vertical
R76 100
Q1 FDY301NZ
BuckPowerGood

BuckAnalogGnd
RenA1 50k
PosBackPlane
RenB1 10k
R140 0
CcC1 12pF 0603
RcB1 28k
CcB1 3.3nF 0603

Pin must be greater than 7.2V to enable buck

Type III compensation network
Used with Ceramic output capacitors

VSw
Rsnub2 0.56 0805, 0.25W
Rsnub 0.56 0805, 0.25W
Csnub2 2.2nF 0603
Csnub 2.2nF 0603

If we supply the boards 20V, we may require a snubber to ensure the switching voltage does not exceed 25V

3V3
R110 100
DP11 LED, 0603, Red Vertical
MON_I2C

U15
ALERT VS
A1 VIN+
A0 VIN-
SDA BUS
SCL GND
Pad
Pins 6,7,8,14,15,16 NC
INA230AIRGTT

I2C
SDA
SCL

Rs1 1k VSw
Rs2 1k 3v3
Cavg 10uF, 0805
C236 0.1uF

3V3
R1 DNL
R2 DNL
R3 10k
R4 10k
A0
A1

The regulator has a soft-start feature which has majority of the control for the first 2.5mS.

Vin=16V, 93% @ 4A, 93% @ 8A, LIR=19%,
4A step ->13mV deviation
0.2W core+winding loss in inductor at 8A

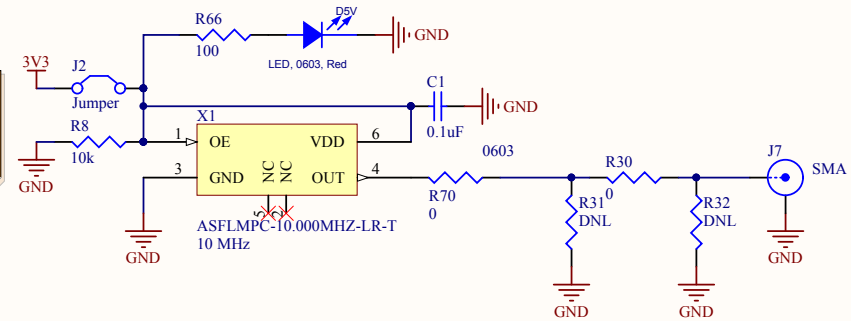
I2C current/voltage sensor.

Unlike the FMC sensors, we use the inductor resistance instead of a precision shunt resistor. Power loss is too high otherwise. LAYOUT: Use Kelvin connection.

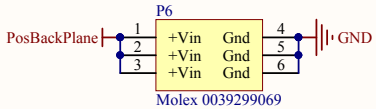
Title: Buck Converter		
Size: Letter	Number	Revision: Rev 0
Date: 4/28/2014		Sheet15of 17
File: C:\Users\..\BuckRegulator.SchDoc		Drawn By: Adam Gilbert

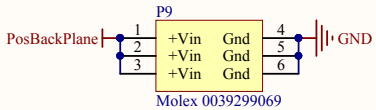
This oscilator is for day 1 testing only

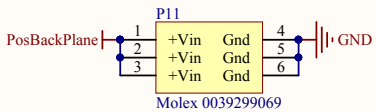
Totally okay to use an SMA cable to connect this to the clock fanout IC

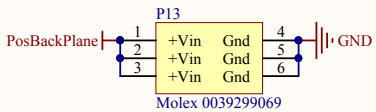


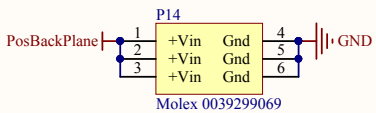
Title: Oscillator		
Size: Letter	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet16of 17
File: C:\Users\..\Oscillator.SchDoc		Drawn By: Adam Gilbert

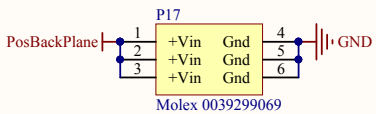


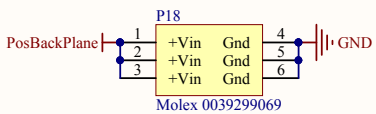


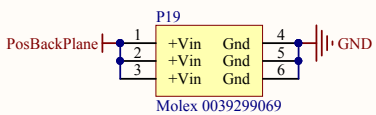


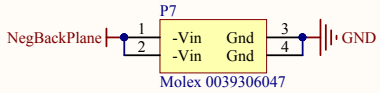


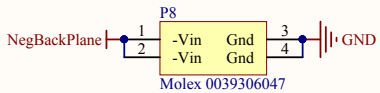


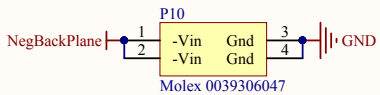


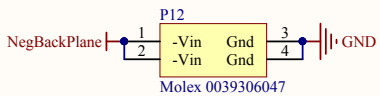


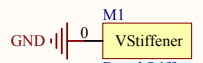












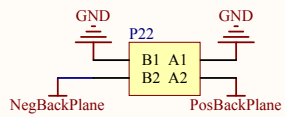
Will not install board stiffener for first build



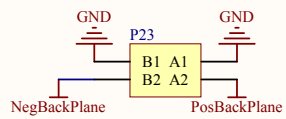








MOLEX IMPACT 78212-1001



MOLEX IMPACT 78212-1001

Title: Mechanical		
Size: Letter	Number:	Revision: Rev 0
Date: 4/28/2014		Sheet 17 of 17
File: C:\Users\..\Mechanical.SchDoc		Drawn By: Adam Gilbert